Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

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Abstract

Chain-of-thought (CoT) prompting is often expected to help reasoning in any language. We test this on date and time arithmetic with Sarvam-105B, using 6,750 questions from the Temporal Reasoning Dataset in English and Hindi. CoT helps English a great deal (+29.5 points) and Hindi very little (+4.7), so the gap between the two languages grows from about 4 points to about 29. Most of this gap follows the language the model reasons in rather than the language of the question: asking for English reasoning on the same Hindi questions closes about four-fifths of it. A two-turn cross-lingual prompt closes about a third, and a Hinglish prompt closes none on Hindi questions while making English questions worse. Two further results are negative. A natural measure of how differently the model reasons in the two languages mostly re-measures whether the final answers differ. Distractor sentences do little harm on average, apart from one sentence, and our resampling experiments cannot separate a distractor from the model's mention of it. Everything comes from one model with one prompt per condition, so we report patterns and do not claim mechanisms.

1 Introduction

Chain-of-thought (CoT) prompting asks a language model to write intermediate steps before answering, and it can improve multi-step reasoning (Wei et al., 2022; Kojima et al., 2022). Whether the benefit carries across languages is less settled. Work on MGSM found English intermediate steps as good as or better than steps in the question's language (Shi et al., 2023b). If that holds, a model asked to reason in Hindi may do worse even when it reads the Hindi question well.

We test this on temporal reasoning using the Temporal Reasoning Dataset (TRD) of Mazzia et al. (2026), which turns CoT off and leaves its effect to future work. TRD can also prepend a distractor sentence to any question, which lets us ask whether reasoning out loud gives irrelevant text more room to do harm, as irrelevant sentences do on English arithmetic (Shi et al., 2023a).

We use Sarvam-105B, whose developer reports giving a substantial part of the training budget to the ten most-spoken Indian languages (Sarvam AI, 2026), and ask five questions:

RQ1 Does CoT help, and does that depend on language and difficulty?

RQ2 Does CoT make distractors more harmful?

RQ3 How much of the English-Hindi gap follows the language of reasoning rather than the language of the question?

RQ4 Does divergence between English and Hindi traces track whether the two languages disagree?

RQ5 Do written mentions of a distractor cause errors?

In short, CoT helps English far more than Hindi, and switching only the reasoning language to English recovers most of the difference. The other three answers are more cautious. Distractor harm is small and concentrated in one sentence; trace divergence predicts disagreement mainly because it contains the answers; and removing a distractor mid-trace helps only a little. Table 1 lists each answer with its headline number.

Our contributions are: (1) to our knowledge, the first CoT evaluation on TRD, split by distractor condition; (2) a decomposition of the gap into reasoning and question language, supported by trace structure; (3) a demonstration that a natural cross-lingual divergence score is largely mechanical, with a way to separate the mechanical part; and (4) on-policy resampling experiments on distractor mentions, with their limits stated. We recomputed every reported number from raw responses and corrected our earlier drafts where they overclaimed.

2 Related Work

CoT and when it stops helping. Wei et al. (2022) introduced CoT with worked examples, and Kojima et al. (2022) showed a trigger phrase alone can work, which suits TRD because it has no worked solutions. Sprague et al. (2025) find CoT helps mainly on math and symbolic problems, so date arithmetic should benefit. Liu et al. (2025) show it can lower performance on some tasks, so a benefit on every task cannot be assumed.

Distractors. GSM-IC adds one irrelevant sentence to GSM8K problems, and accuracy falls under several prompting methods (Shi et al., 2023a). GSM-Symbolic shows sensitivity to clauses that look relevant but are not (Mirzadeh et al., 2025). Neither compares CoT on and off under the same distractor, and both work in English. We use TRD's sentences, and we show below that at least one is not irrelevant in the GSM-IC sense.

Language of reasoning. MGSM compares native and English CoT with few-shot examples on ten languages that do not include Hindi (Shi et al., 2023b). Cross-lingual prompting (CLP) adds an understanding turn before the solving turn (Qin et al., 2023), and cross-lingual thought prompting reports over 10 points of average improvement on arithmetic reasoning and open-domain question answering (Huang et al., 2023). Thinking models often revert to English, and forcing the user's language can cost accuracy (Qi et al., 2025); such models may also reason better in high-resource languages (Yong et al., 2025). Interpretability work suggests English-leaning intermediate processing in other models (Wendler et al., 2024; Zhao et al., 2024); it motivates our question but says nothing direct about Sarvam-105B. Hinglish is an established code-switching pair (Khanuja et al., 2020; Aguilar et al., 2020), and code-mixed examples can help on some tasks (Shankar et al., 2024), though these studies use code-mixed input and do not test generated reasoning.

Traces and faithfulness. A trace need not record the model's computation (Turpin et al., 2023), so faithfulness is tested by intervening on it (Lanham et al., 2023). Thought Anchors resamples continuations to estimate each sentence's counterfactual impact (Bogdan et al., 2025), and Thought Branches argues that hand edits are off-policy and, in decision tasks, have smaller and less stable effects than resampling (Macar et al., 2026). Models can also keep following an injected thought (Yang et al., 2025). Step quality has been scored directly (Golovneva et al., 2023; Prasad et al., 2023); our divergence score is structural and does not assess validity.

Temporal benchmarks. TimeBench and related work evaluate temporal reasoning mostly in English (Chu et al., 2024; Wang and Zhao, 2024). TRD adds ten languages and controlled distractors (Mazzia et al., 2026). We did not find a study crossing CoT, distractors and reasoning language on the same problems, though our search was not systematic.

3 Experimental Design

Data. TRD has nine tasks: date addition, subtraction and duration, time addition, subtraction and duration, date recurrence, interval date and day of week. We generated 150 questions per task at each of five difficulty levels, 6,750 in all, in English and Hindi with matching IDs. Each question appears three times: with no distractor, with a time-related one and with an unrelated one, drawn from pools of 35 and 43 sentence pairs. We recomputed all 13,500 gold answers with Python's `datetime` and all match.

Prompting conditions. All runs use temperature 0, thinking off, a 4,096-token limit and a fixed answer line so responses can be scored automatically. Figure 1 shows one question going through every condition.

Direct asks only for the answer.

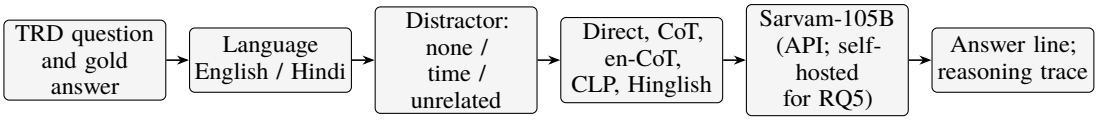
CoT asks the model to reason step by step in the question's language (zero-shot, one stage).

en-CoT keeps the Hindi question and asks for reasoning in English, a zero-shot version of Shi et al. (2023b).

CLP uses the two turns of Qin et al. (2023): first restate and understand the Hindi problem in English, then solve it. They tune temperature per stage where we fix it at 0, we add a rule to write in English, and we do not run their voting variant (CLSP).

Hinglish asks for reasoning that mixes Hindi and English in Roman script, on both question languages.

Direct and CoT run on all 6,750 questions in both languages. The other three run on the first 50 questions of each task and level (2,250 questions). A



Accuracy, drop and penalty; trace length, verification, distractor mention; op tokens and edit distance (RQ4); resampled continuations (RQ5)

English: Today is 2027-05-16, and I have a recurrence every 13 days. Without counting today, what is the date of the 7 occurrence? **Gold:** 2027-08-15.
Hindi: आज 2027-05-16 है, और मेरे पास हर 13 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 7वीं आवृत्ति की तिथि क्या है?
Time-related distractor, prepended: The last time I had a weekly commitment was in January. / आखिरी साप्ताहिक प्रतिबद्धता जनवरी में थी।

Condition	Reasons in	No distractor	Time-related	Condition	Reasons in	No distractor	Time-related
English direct	–	2027-06-05 ×	2027-04-19 ×	en-CoT (Hindi Q.)	English	2027-08-15 ✓	cut off at limit
English CoT	English	2027-08-15 ✓	cut off at limit	CLP (Hindi Q.)	English	2027-08-02 ×	2027-08-02 ×
Hindi direct	–	2027-07-27 ×	2027-03-06 ×	Hinglish (English Q.)	Roman mix	2027-08-02 ×	2027-08-01 ×
Hindi CoT	Hindi	2027-08-02 ×	2027-04-20 ×	Hinglish (Hindi Q.)	Devanagari mix	2027-08-02 ×	2027-07-06 ×

Figure 1: The pipeline, with real responses to one question (long, date recurrence, ID 12). This item was picked to illustrate the design, not as a typical case. The answer 2027-08-02 adds six intervals where seven are needed; the Hindi CoT and CLP traces do so by counting today as the first occurrence. With the distractor, Hindi CoT writes जनवरी 2027 की आखिरी तारीख (31 जनवरी 2027) ही हमारी शुरुआती तारीख है ("the last date of January 2027 (31 January 2027) is our start date"), and English CoT and en-CoT keep trying to fit January into the question until the token limit. Cut-offs count as wrong only under strict scoring.

small pilot with thinking mode on (10 questions per cell, `reasoning_effort` low, 16,000 tokens) gives one more comparison.

Scoring. A response with no answer line usually means the model hit the token limit. *Answered* accuracy leaves such responses out, and our figures use it. *Strict* accuracy counts them as wrong, which is closer to TRD's own scoring (Mazzia et al., 2026). English CoT has the most cut-offs among thinking-off conditions, so answered accuracy flatters it; we give both for headline results and say when a conclusion changes. A distractor's *drop* is accuracy without it minus accuracy with it, and the CoT *penalty* is how much larger that drop is under CoT than under direct answering. For each trace we also record its length, whether it contains a verification phrase (found by regex), and whether it mentions the distractor, compared with a placebo rate on the same question without the distractor.

Comparing traces. For RQ4 a regex parser turns each trace into a sequence of operation tokens with their operands, such as `SUB 2025-05-26 4` for "4 days before 26 May 2025". The reasoning divergence score (RDS) between two traces is the Levenshtein distance (Levenshtein, 1966) between their sequences divided by the longer length, so 0 means the same steps in the same order and 1 means no step in common. Sequences of different lengths cannot match fully, so RDS has a floor set by the lengths alone; Appendix B gives the floor-adjusted version. The parser covers 95–96% of CoT segments on standard replies and 86–88% on thinking-mode reasoning. On 240 tokens read by one LLM annotator, not validated by a person, it gets the operation right 85% of the time and the operation with its arguments 77%.

Statistics. Accuracy intervals are 95% bootstrap intervals (Berg-Kirkpatrick et al., 2012) that resample questions as clusters, since a question appears in three distractor conditions. RQ4 odds ratios use question-clustered standard errors, and task-by-level statements use uncorrected intervals. Task-level tests use exact McNemar with Holm correction over nine tasks (McNemar, 1947; Holm, 1979; Dror et al., 2018), and mention associations use Mantel-Haenszel odds ratios (Mantel and Haenszel, 1959). These intervals treat the distractor sentences and prompts as fixed, so they understate uncertainty about other wordings.

4 Results

4.1 RQ1: CoT helps English far more than Hindi

CoT helps English by about 30 points and Hindi by about 5. Over all 6,750 questions, CoT raises English accuracy by 29.5 points and Hindi by 4.7 (Table 3). The gap between the languages is 4.2 points when the model answers directly and 29.1

	Short answer	Key number
RQ1	CoT helps English much more than Hindi.	+29.5 vs +4.7 points
RQ2	Distractors cost Hindi CoT slightly more than direct answers, largely through one sentence.	Penalty 2.2 points
RQ3	The gap mostly follows the language of reasoning.	23.5 of 29.5 points
RQ4	Trace divergence tracks disagreement mostly through answer tokens.	OR per SD 6.47 $\rightarrow$ 1.14
RQ5	Removing the distractor mid-trace helps a little; it cannot be separated from the mention.	+5.3 points

Table 1: Short answers to the five research questions. Each is explained in its subsection below.

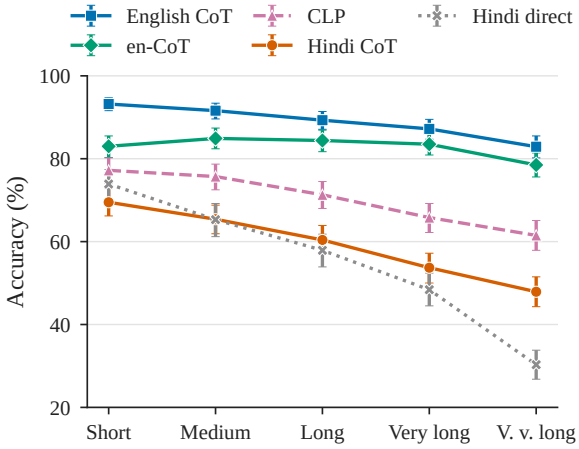


Figure 2: Answered accuracy by difficulty on 2,250 questions, with 95% bootstrap intervals.

with CoT (strict 4.2 and 26.7). It is the same with and without distractors, so pooling does not create it.

For Hindi, CoT can make things worse. Even with no distractor, Hindi CoT lowers accuracy on date subtraction by 19.9 points [16.4, 23.4] and on interval date by 23.3 [19.0, 27.7]. English CoT is never significantly below direct answering on any task, or on any task at any level.

On all 6,750 questions, the gap under CoT grows with difficulty, from 24.0 points on the shortest questions to 34.9 on the longest, but a logistic trend test is not significant ($p = 0.21$), so we describe the widening only in points. Two pooled task and level effects disappear when split by distractor condition (Appendix B).

4.2 RQ2: Distractors do little harm, mostly through one sentence

On average the CoT penalty is small. A time-related distractor lowers Hindi CoT accuracy by 2.6 points [1.3, 3.8], which is 2.2 more than it lowers Hindi direct answers. Removing any single sentence leaves this penalty between 1.8 and 2.6. When the 45 task-sentence units are treated as the sample, however, the interval reaches zero (+2.3 [−0.1, 4.6]), so the penalty may not extend to new sentences. Whether Hindi is penalised more than English depends on how unanswered responses are scored (Appendix B). We therefore withdraw our mid-report claim that distractors make CoT fragile only in Hindi.

One sentence does most of the damage. The largest effect, found post hoc among 90 sentences, comes from the sentence in Figure 1: "The last time I had a weekly commitment was in January." On date recurrence it makes the model try to fit January into the question. English CoT runs to the token limit on 92 of 134 questions (4 of 134 without it), and strict English accuracy on those questions falls from 88.8% to 19.4%. Hindi has no cut-offs but falls from 26.9% to 2.2%. This sentence competes with the start date in the question, so it is not irrelevant in the GSM-IC sense.

Hindi traces mention distractors more often. For time-related sentences, 25.0% of Hindi traces mention the distractor against 8.5% of English traces, with a near-zero placebo rate on a word list chosen after seeing the data. Traces that mention a time-related distractor are less often correct (Hindi OR 0.77 [0.67, 0.89]), while unrelated sentences show no such link. RQ5 tests whether the mention is a cause.

4.3 RQ3: The gap follows the language of reasoning

Switching only the reasoning language to English closes most of the gap. On the same 2,250 questions, English questions with English CoT score 88.9%, Hindi questions with English reasoning (en-CoT) 82.9%, and Hindi questions with Hindi CoT 59.4% (Table 2, Figure 2). Of the 29.5-point gap, 23.5 points lie between en-CoT and Hindi CoT, where only the reasoning language changes. That is 79.7% of the gap, or 80.7% under strict scoring. The remaining 6.0 points lie between the question languages, close to the 4.5-point gap under direct answering, where no reasoning is written (Figure 3). The reasoning part is larger on every task and in every distractor condition.

The traces split the same way. English CoT and en-CoT traces, which share a reasoning language,

	English Q.		Hindi Q.				Hinglish prompt	
	Direct	CoT	Direct	CoT	CLP	en-CoT	English Q.	Hindi Q.
Date addition	85.6	98.0	82.8	85.1	92.7	95.7	87.3	85.9
Date subtraction	86.0	95.2	84.8	66.0	86.9	94.0	70.7	71.0
Date duration	84.0	99.3	78.9	88.2	92.7	97.7	92.9	82.8
Date recurrence	22.1	90.3	2.8	20.1	26.9	70.3	53.8	18.9
Day of week	14.9	69.5	15.2	21.0	31.8	52.5	34.7	23.5
Interval date	43.2	50.7	46.8	23.9	35.3	44.1	38.5	23.6
Time addition	70.5	95.9	75.6	73.5	88.2	92.6	82.9	72.3
Time subtraction	64.7	97.7	64.9	75.2	90.1	95.7	85.2	74.2
Time duration	65.7	98.9	44.7	80.1	86.7	98.1	90.0	80.9
Short	77.8	93.2	73.9	69.5	77.2	83.0	82.9	66.5
Medium	69.6	91.6	65.3	65.4	75.7	84.9	78.8	67.4
Long	61.2	89.3	57.9	60.4	71.3	84.4	70.4	60.9
Very long	54.3	87.2	48.4	53.7	65.8	83.5	64.0	53.7
Very very long	35.5	82.9	30.3	47.9	61.5	78.5	58.2	48.3
No distractor	60.6	89.3	56.0	60.2	69.3	84.1	70.6	61.0
Time-related	59.9	88.2	54.9	58.9	69.3	81.0	69.7	59.1
Unrelated	58.6	89.2	54.6	59.1	72.3	83.6	72.3	58.0
All, answered	59.7	88.9	55.2	59.4	70.3	82.9	70.9	59.4
All, strict	59.6	86.0	55.2	59.0	69.9	80.8	70.3	59.1

Table 2: Answered accuracy (%) on the same 2,250 questions, by task, level and distractor condition. The last row counts unanswered responses as wrong. Differences quoted in the text are computed before rounding, so they can differ from these cells by 0.1.

	Direct	CoT	Gain [95% CI]	Strict
English	59.0	88.5	+29.5 [28.5, 30.6]	+26.8
Hindi	54.7	59.4	+4.7 [3.6, 5.7]	+4.2
By distractor	None		Time-related	Unrelated
English	+28.8		+29.1	+30.7
Hindi	+5.6		+3.4	+5.0
Gap under CoT	+28.0		+29.7	+29.7
Gap, direct	+4.8		+4.0	+3.9

Table 3: CoT gain on all 6,750 questions, in points. All values are answered except the Strict column, which counts unanswered as wrong. By-distractor English and Hindi rows are gains. Gains and gaps are computed before rounding.

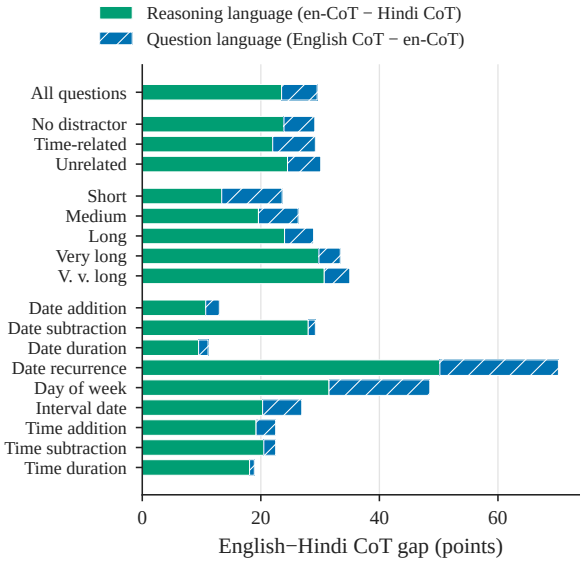


Figure 3: The English-Hindi CoT gap on 2,250 questions, split on one path through three conditions. No condition pairs English questions with Hindi reasoning, so the split is descriptive.

are about as far apart (RDS 0.730) as two English traces of one question with and without a distractor (0.713). en-CoT and Hindi CoT traces on identical Hindi questions are as far apart (0.837) as English and Hindi CoT (0.852), and traces of unrelated questions sit at 0.959 (Figure 4). Self-checking follows the reasoning language too: a verification phrase appears in 86.1% of English CoT traces, 74.2% of en-CoT traces and 1.3% of Hindi CoT traces. Verification does not explain the en-CoT gain, though: an observational estimate within task and level attributes only 1.0% [−6.3, 9.0] of it to verification.

CLP helps less than one English turn. CLP scores 70.3%, 10.9 points above Hindi CoT but 12.6 below en-CoT. This reverses the order Qin et al. (2023) report for GPT-3.5 on MGSM, although model, task and settings all differ. One candidate reason is that the understanding turn often solves the problem already and the solving turn copies it. On the three date-answer tasks, the second turn is right in 97.5% of responses when the first ends at

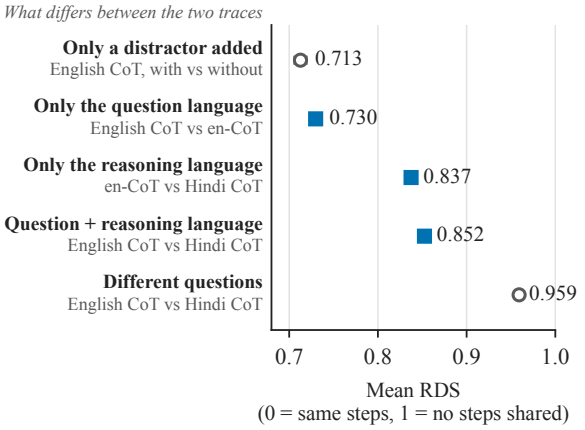


Figure 4: How different two reasoning traces are, as mean RDS with 95% intervals on 2,250 questions. Each row compares two traces that differ in only the factor named in bold; en-CoT is a Hindi question with English reasoning. The middle rows compare traces of the same question. The hollow grey rows are reference points: the smallest change we make (adding a distractor) and traces of unrelated questions.

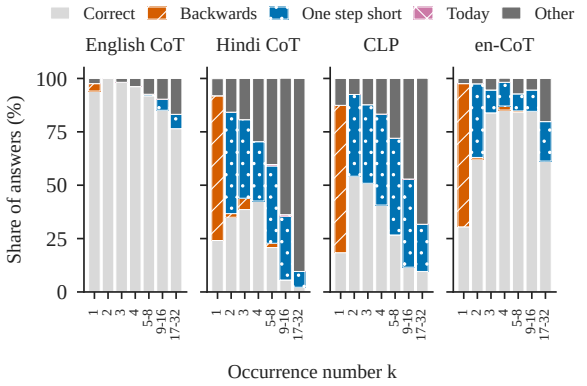


Figure 5: Date recurrence answers by occurrence number k, on 2,250 questions. Backward answers concentrate at $k = 1$ under every prompt on Hindi questions; one-step-short answers run through every k.

the gold date and in 20.1% when it does not. The first turn was not randomised, so this is not proof, and our temperature of 0 and added language rule may also contribute.

Hinglish does not help Hindi and hurts English. Asking for Hinglish reasoning costs English questions 18.0 points [16.8, 19.2] (strict 15.7), on all nine tasks. Hindi questions do not change on average (−0.0 [−1.3, 1.2]), so the gap narrows only because English gets worse. The model also largely ignores the Roman-script instruction on Hindi questions: 68.5% of those traces are mostly Devanagari. The English loss is not only the cost of writing in Hindi. Traces with no Hindi function words still lose 10.3 points [8.3, 12.3], and the share that verify falls from 86.5% to 18.6%. One failure looks like Hindi CoT's: on date subtraction across a month boundary, English questions fall from 90.1% to 42.3%, near Hindi CoT's 36.2% (Appendix A has an example), while same-month questions stay at 97.4%.

One error does not depend on the reasoning language. On date recurrence questions asking for the first occurrence ($k = 1$), about two-thirds of answers to Hindi questions go backwards in time, whether the reasoning is in English (en-CoT 67.1%), Hindi (67.8%) or through CLP (69.0%), against 3.5% for English CoT (Figure 5). The Hindi template's "आज को छोड़कर, 1वीं आवृत्ति" is a candidate cause, since 1वीं is non-standard. Yet equally non-standard ordinals at $k = 2$ to 4 almost never produce backward answers, and we did not rerun a corrected template, so the cause is open.

Thinking mode. With thinking on, Hindi CoT reaches 97.9% answered and 89.9% strict, and its hidden reasoning is 79% English words. This fits the account above but cannot confirm it, because thinking mode also had four times the token budget. Cut at 4,096 tokens, strict Hindi CoT would score 43.8%, below the 59.9% with thinking off on the same pilot questions, and within these traces a higher English share does not predict higher accuracy.

4.4 RQ4: Divergence is mostly built into the metric

At first sight, trace divergence strongly predicts disagreement. On 20,662 answered pairs, English and Hindi disagree on 31.4%, and each 0.1 of RDS multiplies the odds of disagreement by 4.92 [4.58, 5.30], controlling for task, difficulty, distractor and mode.

Most of that link comes from the answers themselves. RDS includes the final-answer tokens, so two traces that reach different answers already differ. Replacing only Hindi's answer values in pairs where both languages are right accounts for 68% of the RDS difference between agreeing and disagreeing pairs. Removing answer tokens drops the odds ratio per SD from 6.47 to 1.65, and comparing only operation types (for example `ADD` against `SUB`, without operands) gives 1.14 (Figure 6). Out of fold, op-type divergence adds 0.0007 AUC [−0.0004, 0.0018] to a model that already

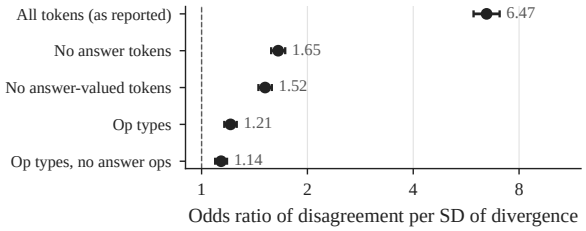


Figure 6: Odds ratio of disagreement per SD of divergence, under task, difficulty, distractor and mode controls, with SEs clustered by question. The sample is the 20,648 answered pairs on which every token view is non-empty, so all rows share it; the main model in the text uses all 20,662.

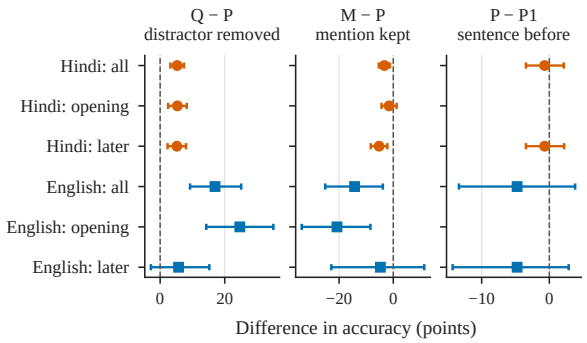


Figure 7: Resampling contrasts with 95% intervals. Traces were selected for being wrong, the distractor is time-related, and unanswered continuations count as wrong. *Opening* traces mention the distractor in their first sentence ($k = 0$) and *later* traces after it ($k \geq 1$); P−P1 uses only later traces. English is descriptive.

knows task, difficulty and distractor, while simply comparing the two final answers gives AUC 0.951.

RDS tracks failure more than disagreement. Among answered thinking-off pairs, RDS is joint highest where both languages are wrong (0.925, the same as where only Hindi is wrong), and those pairs count as agreement. This does not mean failing traces follow the same route: within Hindi, wrong and right traces on one question do differ (+0.030 [0.021, 0.039]). The choice of normalisation also changes one comparison, between thinking-mode and standard traces (Appendix B).

4.5 RQ5: Distractor mentions under intervention

Setup. We take wrong CoT answers to questions with a time-related distractor whose trace mentions the distractor before answering: 526 Hindi and 37 English. Testing whether the mention caused the error needs the model to continue a trace from the middle, which the API does not allow, so we served an FP8 build of the model with vLLM. Against the API it scored 5.7 points higher on 278 Hindi questions and 24.7 points lower on 22 English ones, so English results are descriptive only.

Deleting the mention. Deleting the sentence that mentions the distractor helps no more than deleting a neutral sentence (+0.0 [−6.8, 7.2] on 222 Hindi traces, greedy decoding). With one continuation per version this test had about 24% power for a 5-point effect, so it does not show the effect is zero.

Resampling. Following Bogdan et al. (2025) and Macar et al. (2026), let k be the first sentence that mentions the distractor. We resample seven continuations per trace at temperature 0.8 from four starting points: *P* continues from just before k; *Q* does the same with the distractor removed from the question; *M* continues from just after k; and *P1* from one sentence before k.

Removing the distractor helps a little. In Hindi, Q reaches 45.2% and P 40.0%, a difference of +5.3 points [3.2, 7.4] (Figure 7). So the distractor was necessary for at least 8.8% of the errors made from that point (Tian and Pearl, 2000); most continuations that go wrong with it also go wrong without it. The effect leans on one sentence, again the one from Figure 1, which supplies 58% of it (without it, +2.8 [0.4, 5.0]), and the interval reaches zero once the 29 sentences are treated as a sample.

The mention itself looks like an ordinary step. Keeping the sentence with the first mention lowers accuracy by 3.3 points [1.2, 5.4], but not detectably more than keeping the sentence before it ($p = 0.09$ on 248 traces with $k \geq 1$). It cannot be told apart from one more step along a trace that was heading for a wrong answer.

We cannot separate the distractor from writing about it. Free continuations mention the distractor in 57.8% of rollouts, so the Q-versus-P contrast changes both at once. Unanswered Hindi rollouts also rise from 5.7% to 10.0%, which the API data does not show, so part of this may belong to the self-hosted setting.

5 Discussion

Ask for English reasoning, at least on this model. Across accuracy, trace distance and verification, the instructed language of reasoning accounts for more of the gap than the language of the question. This fits the small direct-answer gap and prior work favouring English reasoning (Shi et al., 2023b;

Yong et al., 2025; Qi et al., 2025). The en-CoT gain here is far larger than MGSM's, which may reflect zero-shot prompting, the task or the model. We did not test replying to users in Hindi after reasoning in English.

Prompting results may not transfer. A single English turn beat two turns here, the reverse of the original CLP result, and carry-over from the first turn is a candidate explanation. A Hinglish instruction made English questions worse, partly without any Hindi in the trace. Those traces verified far less often, but the model chooses whether to verify, so we do not treat verification as the mechanism.

Check trace metrics against the answer. A trace metric that includes the answer will look predictive of whether answers agree. A large odds ratio needs a version without those tokens and a baseline before it can be read. Reading a wrong trace that mentions a distractor can also mislead, since most such errors recur without the distractor.

6 Conclusion

On Sarvam-105B and TRD, CoT widens the English-Hindi gap from about 4 points to about 29. Most of that gap closes when the same Hindi questions are reasoned through in English, a third of it with CLP, and on average none with a Hinglish prompt on Hindi questions, whose traces stayed mostly Devanagari. Divergence between traces predicts disagreement far more weakly once answer tokens are removed. The average distractor effect is small and varies by sentence, and our design cannot separate it from the model's mention of it.

7 Limitations and Future Work

Scope. Everything here concerns one model, greedy decoding except in the resampling arms, one prompt per condition, one dataset seed and two languages. Intervals cover question sampling only, so they say nothing about other wordings, and with temperature 0 there is no sampling noise estimate for the main runs.

Identification. Language and prompt wording are confounded in every comparison across prompting conditions, including our decomposition. No condition pairs English questions with Hindi reasoning, so the split is one path and depends on its order. Trace style, mentions and verification are chosen by the model or measured after it responds, so they support association only. Runs span about a month, the provider changed its API responses in that time, and condition and run date are partly confounded.

Data. The added conditions use the first 50 of 150 questions per cell. This subset is too small for distractor contrasts of a few points. Hindi distractors are not literal translations, and some Hindi carriers are awkward, so English-Hindi distractor comparisons are not strictly parallel. Distractor sentences are a fixed pool assigned unevenly across tasks, and the generator's seed reset makes levels share sentence mixes. TRD's time-task offsets in the code differ from the difficulty ranges its paper lists.

Measurement. The parser is regex-based and imperfect. Mention and verification rules are lexical. All our hand-labelled samples were read by a single LLM annotator without blinding or human validation, including parser precision and our review of the Hindi templates; they should be checked by people.

Interventions. Resampling covers only originally wrong traces and time-related distractors, on a quantised self-hosted model at a different temperature from the source traces. The resampling design was written down before the run according to our notes, but it was first committed afterwards, followed the deletion experiment on many of the same traces, and was not publicly registered. We did not test whether mentions harm traces that would otherwise be correct.

Thinking mode. This pilot has 10 questions per cell and a fourfold budget, and it changes thinking and budget together.

Future work. The most informative next steps are: rerunning recurrence with corrected Hindi carriers; adding English questions with Hindi reasoning to complete the design; an English CoT prompt without the language lock and several paraphrases of each prompt; CLP with solving forbidden in stage 1; thinking on and off at equal budgets; a question-by-mention factorial on both correct and wrong traces; and human annotation of the labelled samples.

References

Gustavo Aguilar, Sudipta Kar, and Thamar Solorio. 2020. LinCE: A centralized benchmark for linguistic code-switching evaluation. In *Proceedings of the*

Twelfth Language Resources and Evaluation Conference, pages 1803–1813, Marseille, France. European Language Resources Association.

Taylor Berg-Kirkpatrick, David Burkett, and Dan Klein. 2012. An empirical investigation of statistical significance in NLP. In *Proceedings of the 2012 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*, pages 995–1005, Jeju Island, Korea. Association for Computational Linguistics.

Paul C. Bogdan, Uzay Macar, Neel Nanda, and Arthur Conmy. 2025. Thought anchors: Which LLM reasoning steps matter? ArXiv:2506.19143.

Zheng Chu, Jingchang Chen, Qianglong Chen, Weijiang Yu, Haotian Wang, Ming Liu, and Bing Qin. 2024. TimeBench: A comprehensive evaluation of temporal reasoning abilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1204–1228, Bangkok, Thailand. Association for Computational Linguistics.

Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker's guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392, Melbourne, Australia. Association for Computational Linguistics.

Olga Golovneva, Moya Chen, Spencer Poff, Martin Corredor, Luke Zettlemoyer, Maryam Fazel-Zarandi, and Asli Celikyilmaz. 2023. ROSCOE: A suite of metrics for scoring step-by-step reasoning. In *The Eleventh International Conference on Learning Representations*, Kigali, Rwanda.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Haoyang Huang, Tianyi Tang, Dongdong Zhang, Wayne Xin Zhao, Ting Song, Yan Xia, and Furu Wei. 2023. Not all languages are created equal in LLMs: Improving multilingual capability by cross-lingual-thought prompting. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 12365–12394, Singapore. Association for Computational Linguistics.

Simran Khanuja, Sandipan Dandapat, Anirudh Srinivasan, Sunayana Sitaram, and Monojit Choudhury. 2020. GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3575–3585, Online. Association for Computational Linguistics.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems*, volume 35, pages 22199–22213.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamilė Lukošiūtė, Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan Kundu, and 11 others. 2023. Measuring faithfulness in chain-of-thought reasoning. ArXiv:2307.13702.

Vladimir I. Levenshtein. 1966. Binary codes capable of correcting deletions, insertions, and reversals. *Soviet Physics Doklady*, 10(8):707–710.

Ryan Liu, Jiayi Geng, Addison J. Wu, Ilia Sucholutsky, Tania Lombrozo, and Thomas L. Griffiths. 2025. Mind your step (by step): Chain-of-thought can reduce performance on tasks where thinking makes humans worse. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 38489–38517. PMLR.

Uzay Macar, Paul C. Bogdan, Senthooran Rajamanoharan, and Neel Nanda. 2026. Thought branches: Interpreting LLM reasoning requires resampling. In *The Fourteenth International Conference on Learning Representations*.

Nathan Mantel and William Haenszel. 1959. Statistical aspects of the analysis of data from retrospective studies of disease. *Journal of the National Cancer Institute*, 22(4):719–748.

Vittorio Mazzia, Sandro Pollastrini, Davide Bernardi, Chiara Rubagotti, and Daniele Amberti. 2026. Benchmarking multilingual temporal reasoning in LLMs: The temporal reasoning dataset. In *Proceedings of the 16th International Workshop on Spoken Dialogue System Technology*, pages 168–181, Trento, Italy. Association for Computational Linguistics.

Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2025. GSM-symbolic: Understanding the limitations of mathematical reasoning in large language models. In *The Thirteenth International Conference on Learning Representations*.

Archiki Prasad, Swarnadeep Saha, Xiang Zhou, and Mohit Bansal. 2023. ReCEval: Evaluating reasoning chains via correctness and informativeness. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 10066–10086, Singapore. Association for Computational Linguistics.

Jirui Qi, Shan Chen, Zidi Xiong, Raquel Fernández, Danielle Bitterman, and Arianna Bisazza. 2025. When models reason in your language: Controlling thinking language comes at the cost of accuracy. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 20279–20296, Suzhou, China. Association for Computational Linguistics.

Libo Qin, Qiguang Chen, Fuxuan Wei, Shijue Huang, and Wanxiang Che. 2023. Cross-lingual prompting: Improving zero-shot chain-of-thought reasoning across languages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2695–2709, Singapore. Association for Computational Linguistics.

Sarvam AI. 2026. Open-sourcing Sarvam 30B and 105B. Blog post. Accessed: 2026-10-02.

Bhavani Shankar, Preethi Jyothi, and Pushpak Bhattacharyya. 2024. In-context mixing (ICM): Code-mixed prompts for multilingual LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4162–4176, Bangkok, Thailand. Association for Computational Linguistics.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. 2023a. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227. PMLR.

Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2023b. Language models are multilingual chain-of-thought reasoners. In *The Eleventh International Conference on Learning Representations*, Kigali, Rwanda.

Zayne Rea Sprague, Fangcong Yin, Juan Diego Rodriguez, Dongwei Jiang, Manya Wadhwa, Prasann Singhal, Xinyu Zhao, Xi Ye, Kyle Mahowald, and Greg Durrett. 2025. To CoT or not to CoT? chain-of-thought helps mainly on math and symbolic reasoning. In *The Thirteenth International Conference on Learning Representations*.

Jin Tian and Judea Pearl. 2000. Probabilities of causation: Bounds and identification. In *Proceedings of the Sixteenth Conference on Uncertainty in Artificial Intelligence*, pages 589–598.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36.

Yuqing Wang and Yun Zhao. 2024. TRAM: Benchmarking temporal reasoning for large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 6389–6415, Bangkok, Thailand. Association for Computational Linguistics.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837.

Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Do llamas work in English? on the latent language of multilingual transformers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15366–15394, Bangkok, Thailand. Association for Computational Linguistics.

Sohee Yang, Sang-Woo Lee, Nora Kassner, Daniela Gottesman, Sebastian Riedel, and Mor Geva. 2025. How well can reasoning models identify and recover from unhelpful thoughts? In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 7030–7047, Suzhou, China. Association for Computational Linguistics.

Zheng-Xin Yong, M. Farid Adilazuarda, Jonibek Mansurov, Ruochen Zhang, Niklas Muennighoff, Carsten Eickhoff, Genta Indra Winata, Julia Kreutzer, Stephen H. Bach, and Alham Fikri Aji. 2025. Crosslingual reasoning through test-time scaling. Computing Research Repository, arXiv:2505.05408.

Yiran Zhao, Wenxuan Zhang, Guizhen Chen, Kenji Kawaguchi, and Lidong Bing. 2024. How do large language models handle multilingualism? In *Advances in Neural Information Processing Systems*, volume 37.

Appendix A. Worked Examples

Hinglish, English question, wrong. Date subtraction, short. "Today is 2025-05-01, what was the date 3 days ago?" Gold 2025-04-28. Standard English CoT answered correctly. Excerpt: *Step 4: 01 minus 3 karne par hum 31 April par aayenge. Step 5: Isliye, 3 days ago ki date 2025-04-30 thi.* Answer: 2025-04-30.

en-CoT, $k = 1$, backward answer. Date recurrence, short, ID 5, time-related distractor, Hindi question with आज को छोड़कर, 1वीं आवृत्ति. Excerpt: *Given the phrasing "excluding today," this suggests we need to find what comes before.* Answer: 2025-04-06. Gold 2025-04-14.

CLP, stage 1 anchors stage 2. Date recurrence, long, ID 12, no distractor. Stage 1 ends: *Therefore, the date of the 7th occurrence is 2 August 2027.* Stage 2 ends: *Adding 78 days to 16 May 2027 also results in 2 August 2027.* Answer: 2027-08-02. Gold 2027-08-15.

Resampling arms. Date recurrence, long, ID 12, Hindi, time-related distractor. The original trace answered 2027-04-20. Correct continuations out of seven: Q 5, P 2, M 0, P1 2. A single example cannot show an effect; Figure 7 gives the aggregates.

Appendix B. Further Statistics

Method details. A substitution between a partial and a full date costs 0.5 in RDS, a rule whose effect we confirmed is negligible. The length floor is $|n - m| / \max(n, m)$, and RDS equals floor plus $(1 - \text{floor})$ times a floor-adjusted score. On the 2,250-question subset the 29.5-point gap has interval [28.1, 30.9], the reasoning part 23.5 [22.1, 24.9], the question part 6.0 [5.0, 7.1], and the reasoning share 79.7% [76.4, 82.9].

RQ1. The time duration gain gap (English gain minus Hindi gain), +2.3 pooled, is −5.2 with no distractor, +14.5 with a time-related one and −2.2 with an unrelated one; only +14.5 is significant. Hindi CoT's harm at the short level (−4.3 pooled) comes from the distractor conditions (−1.9 [−4.5, 0.7] without one, −6.2 and −4.7 with). Both Hindi task losses in RQ1 stay significantly negative with either distractor. Under strict scoring the gap under CoT grows from 23.0 to 31.1; the logistic trend has OR 0.97 per level.

RQ2. The Hindi direct-to-CoT penalty for time-related sentences has interval [0.6, 3.8]. Comparing languages, the Hindi-minus-English penalty is +2.6 [0.4, 4.7] answered but +1.4 [−0.7, 3.5] strict, as English CoT's drop grows from 0.9 to 2.2 [1.3, 3.2]; it is not significant on logit specifications unless recurrence is removed. With unrelated sentences it is +2.6 [0.5, 4.6] answered and +2.8 [0.8, 4.8] strict, but because English CoT resists a fall in English direct answers: the Hindi penalty itself is null (+0.7). Of the 92 cut-off traces under the January sentence, none is a verbatim loop, and in a systematic sample of 23, read from excerpts by one LLM annotator, 18 mainly search for a January start date. Removing that sentence takes the Hindi recurrence penalty from +4.5 to +0.3 [−4.6, 4.9]; English stays at +8.5 (strict +9.9). For mentions, the English OR on the stricter list is 0.57 [0.38, 0.85] (0.92 on the original list); unrelated sentences give 1.15 and 1.30.

RQ3. The reasoning part is 22.0 to 24.5 points in each distractor condition, and the two parts are not distinguishable at the short level. The direct-answer gap is 4.5 [3.3, 5.8]. Verification is associated with 63.6% [39.8, 97.0] of en-CoT's day-of-week residual to English. CLP gains +9.0, +10.4 and +13.3 over Hindi CoT by distractor condition, and beats Hindi direct by 15.1 [13.5, 16.7], but is significantly worse on interval date (35.3% vs 46.8%). en-CoT beats CLP by 12.6 [11.4, 13.9]; on MGSM Qin et al. report CLP 70.6 and En-CoT 57.8. Within the same question, the stage-1 split differs by +62.5 points [57.1, 68.0], against about 7 for the same split on en-CoT; Qin et al. also note that stage 1 may give preliminary answers. For Hinglish, date duration falls by 5.5 points and date subtraction rises by 5.1 on Hindi questions; 25.5% of Hindi-question traces have no Devanagari. Traces with Hindi function words lose 11.3 points more [9.0, 13.6] within task, level and distractor (8.4 to 12.0 across 48 threshold pairs). On Hindi questions Roman traces do no better than Devanagari ones (−2.8 [−5.2, −0.5]; −2.3 [−5.0, 0.4] within question). For recurrence, non-standard ordinals at $k = 2$ to 4 give backward answers on 2 of 229 en-CoT questions, and removing $k = 1$ moves en-CoT's gap to English only from 6.0 to 5.3. en-CoT's gap to English narrows from 10.2 points at short to 4.3 at very very long, but without $k \leq 2$ the change is +0.8 [−2.5, 4.1].

RQ4. The answer-swap check uses the 11,309 both-correct thinking-off pairs and raises mean RDS by 0.082, against a 0.120 gap to disagreeing pairs. With op types including answer ops the OR per SD is 1.21. The fall in OR per 0.1 appears in each distractor condition (5.37, 4.60, 4.95 to 1.13, 1.08, 1.09). Within a question, op-type divergence barely separates disagreeing from agreeing variants (concordance 0.516 [0.501, 0.530]), and a within-question slope survives Holm correction only for date recurrence (OR per 0.1, 1.42 [1.25, 1.61]) and day of week (1.26 [1.11, 1.43]). Op types separate a different question from the same question by only +0.012. The floor is 58–61% of mean RDS on standard content and on thinking-mode reasoning. Raw RDS does not separate the two on matched questions (+0.005 [−0.002, 0.011]), but the floor-adjusted score does (+0.088 [0.072, 0.105]). Counting unanswered as wrong on the same 450 questions, disagreement is 10.9% with thinking on against 31.7% off (OR 0.23 [0.18, 0.30]), with thinking and budget changing together.

RQ5. Agreement with the API on 90 calibration questions is −2.2 points [−13.3, 7.8], which does not establish that the two are the same; the Hindi difference on 278 questions is [0.5, 10.8]. Deletion with sampling gives +4.5 [−2.7, 11.7]. The net 8.8% has interval [5.5, 12.1], and resampling the 29 sentences widens the Q−P interval to [−0.3, 9.4]. The mention test is (M−P)−(P−P1) = −4.6 [−9.6, 0.6], with P−P1 −0.7 [−3.5, 2.1]. In later-mention traces, rollouts with no mention match the no-distractor arm (+0.0 [−4.8, 4.7]) and those with a mention do not (+8.7 [3.7, 13.6]); these subgroups are post-treatment. On the API, Hindi non-completion barely moves with the distractor (0.81% to 0.89%).

Appendix C. Secondary Figures

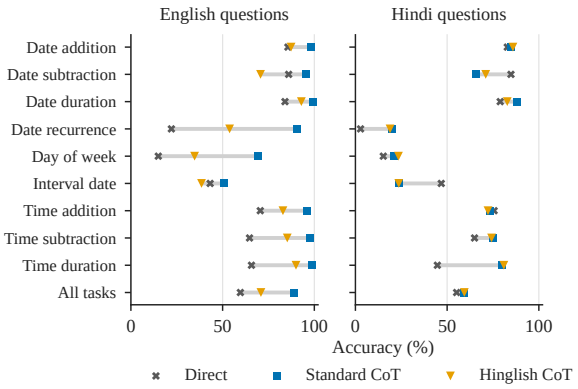


Figure 8: Accuracy by task under the Hinglish prompt, answered. Standard CoT is the CoT condition, reasoning in the question's language.

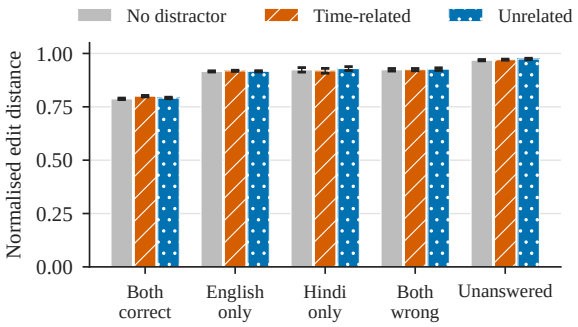


Figure 9: Mean RDS by correctness outcome and distractor condition. Failure outcomes sit about 0.13 above both-correct.